

Problem Solving with Volume

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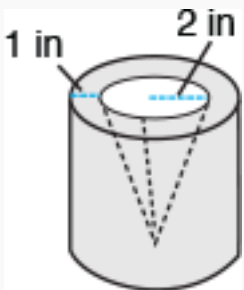
Problem Solving with Volume

You can combine solids to form more complex figures.

Example

Hide

A cone shape is cut from a foam cylinder to create a form used to dry painted machine parts. If the height of the cylinder is 6 inches and the height of the cone is 5 inches, how many cubic inches of foam are left in the form? Round to the nearest cubic inch. (**Hint:** The radius of the cylinder is the radius of the cone plus the 1-inch thickness of the foam.)



Step 1

Find the volume of both figures.

Cylinder: $V = \pi r^2 h$	Cone: $V = \frac{1}{3} \pi r^2 h$
$= 3.14 \cdot 3^2 \cdot 6$	$= \frac{1}{3} \cdot 3.14 \cdot 2^2 \cdot 5$
$\approx 169.56 \text{ in}^3$	$\approx 20.93 \text{ in}^3$

Step 2

Subtract and round to the nearest cubic inch.

$169.56 - 20.93 = 148.63$, which rounds to 149 in³.

There are **149 in³** left in the foam.

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^ High

Medium

Low

Category

Lesson 4: Volume

 Related Questions:

Practice questions taken related to this topic